

Assignment 4: Data pipeline with Airflow

Learning Objectives:

In this exercise, you will learn how to build a data pipeline using Apache Airflow. More specifically, you will create data pipe to perform ETL on a remote data source at the following location:

```
URL = 'https://elasticbeanstalk-us-east-2-340729127361.s3.us-east-2.amazonaws.com/trafficdata.tgz'
```

Due Date: February 16, 2025

Evaluation: 10% of the total grade

Instructions

You are provided with a link to a remote data source for this assignment. There are three data files in the zip folder with details in the fileformats.txt file. You will do the following to accomplish this:

Create a “dags” folder where you have initiated the standalone server and write your DAG in this folder with the following arguments:

Owner='your_name', dag_id='traffic_data_ETL', start_date=datetime(2025, 2, 10), schedule='@daily'

Your DAG should perform the following tasks :

1. Define a task to create a directory structure to save files throughout the pipeline run
2. Define a task to download and save the zip folder
3. Define a task to unzip and save the data
4. Define a task to extract data from the CSV file. Cut the columns 1, 2, 3, and 4 to extract Rowid, Timestamp, Anonymized Vehicle number, and Vehicle type from the vehicle-data.csv file and save it to csv_d.csv file.
5. Define a task to extract data from the TSV file. Cut the columns 5, 6 and 7 to extract Number of axles, Tollplaza id, and Tollplaza code from the tollplaza-data.tsv file and save it as tsv_d.csv.
6. Define a task to extract data from the fixed width file. Specify the start and end of each column to extract Type of Payment code, and Vehicle Code from the fixed width file payment-data.txt and save it into a file named fixed_width_d.csv
7. Define a task to combine all three extracted files from steps 4-6 above (paste command) and save the combined data file
8. Define a task to transform the data by converting “Vehicle type” (column 4) to uppercase letters and save the transformed data in a CSV file
9. Define the pipeline dependencies (hint: tasks 4-6 can run in parallel)
10. Start the Airflow standalone server by using command “airflow standalone”
11. Switch to airflow local server at URL = <http://localhost:8080>. Your DAG should appear in there.
12. Activate and trigger your DAG, and explore the pipeline by examining Graph, Logs, Gant, etc.

Note: Submit your code and report with embedded screenshots to substantiate your work.

Scoring Rubric

Category	Criteria	Scores
Pipeline	Successful implementation of the DAG showing all tasks in the Airflow UI, runs on manual trigger, completes all tasks successfully, data output as expected, Graph & schedule as per specifications	4
Python code	Well organized and clean code with appropriate comments, implementing DAG performing all the tasks	4
Report	A well-organized report containing an executive summary followed by properly labeled sections showing results, screenshots and brief explanation of observations	2